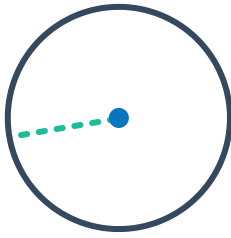


Features of the circle

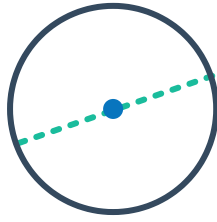


1 Name the indicated part of the following circles:

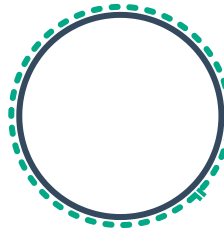
a



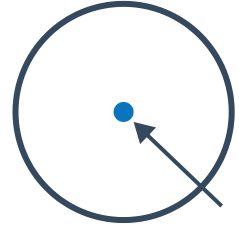
b



c

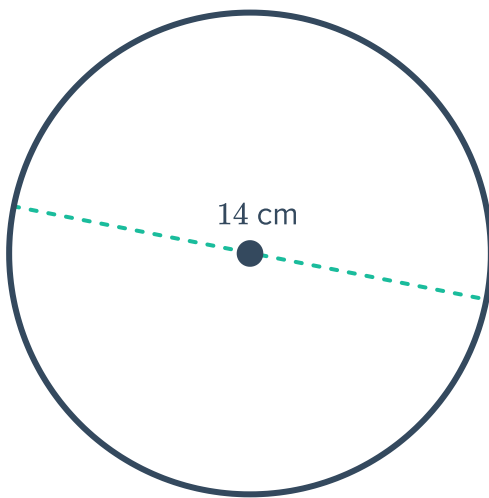


d

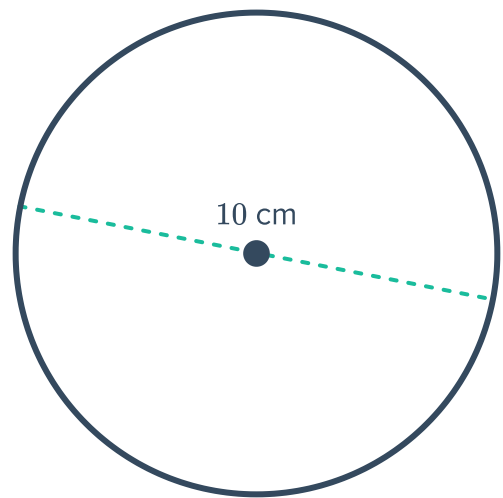


2 Find the radius of the following circles:

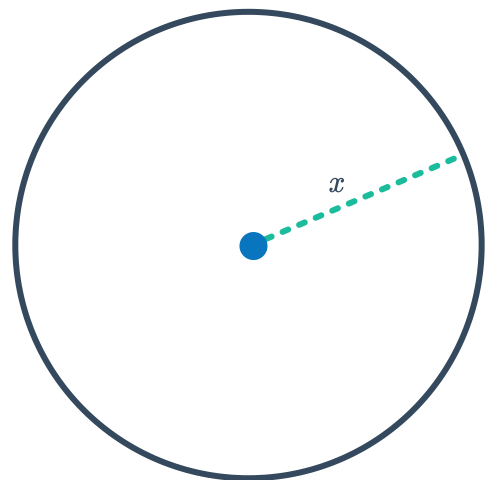
a



b

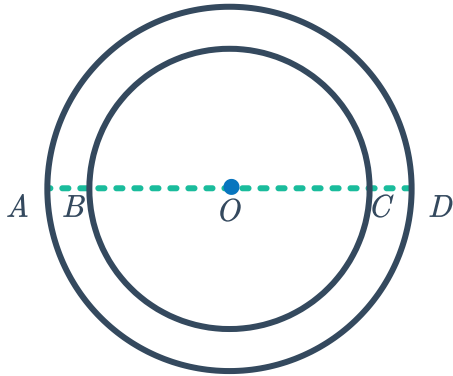


3 Find diameter of the following circle:



4 The radius of the smaller circle shown in the figure is 16 cm:

If $CD = 4$ cm, find AD , the diameter of the larger circle.



5 Complete the following tables:

a

	Diameter (cm)	Radius (cm)
Circle 1		11
Circle 2		17
Circle 3		19
Circle 4		21
Circle 5		23

b

	Diameter (cm)	Radius (cm)
Circle 1	22	
Circle 2	26	
Circle 3	28	
Circle 4	36	
Circle 5	42	

Circumference

6 Determine whether the following statements are true about π .

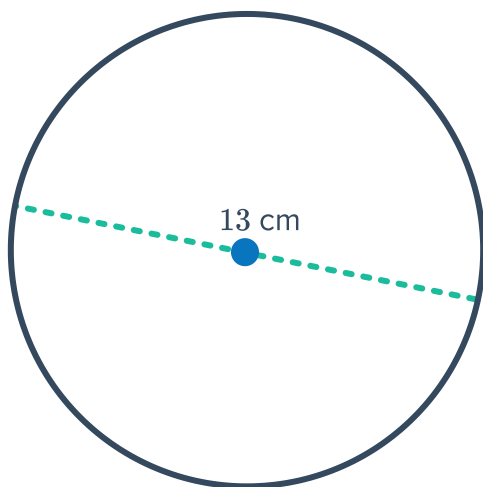
- a π is equal to 3.142.
- b π cannot be expressed as a fraction using whole numbers.
- c π is the ratio between the circumference and the diameter of a circle.
- d π is equal to $\frac{22}{7}$.
- e π is the ratio between the circumference and the radius of a circle.
- f π is equal to 3.14



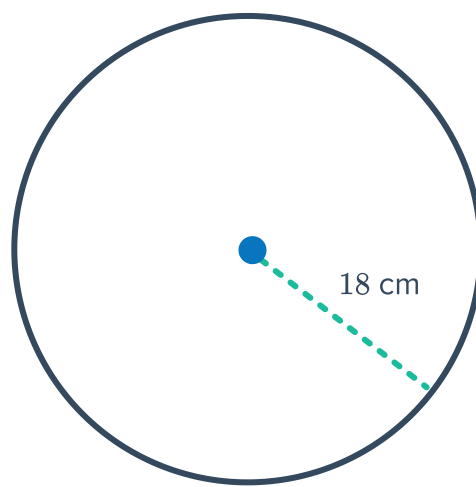
7 For each of the following circle:

- i Calculate the exact circumference in terms of π .
- ii Hence, calculate the circumference rounded to two decimal places.

a



b



c A circle with a diameter of 7.7 cm.

d A circle with a radius of 4.1 cm.

8 Calculate the exact circumference of the following circles:

a A circle with a diameter of 59 cm.

b A circle with a radius of 22 cm.

9 Calculate the circumference of the following circles, correct to two two decimal places:

a A circle with a diameter of 38 cm.

b A circle with a diameter of 45 cm.

c A circle with a radius of 27 cm.

d A circle with a radius of 1.8 cm.

10 Caitlin and David calculate the circumference of this circle using different formulas.

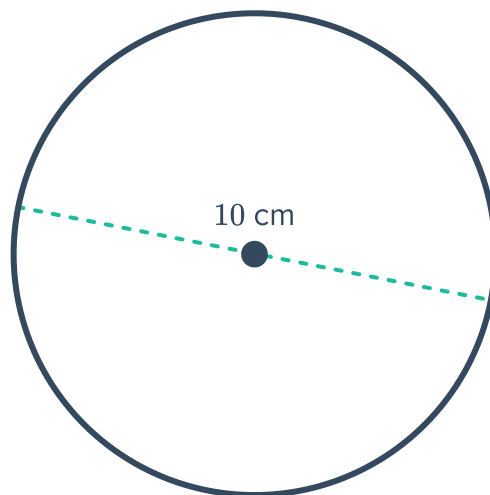
a Caitlin uses the formula $C = \pi d$ to calculate the circumference.

Find her result as an exact value

b David uses the formula $C = 2\pi r$ to calculate the circumference, where r is the radius.

Find his result as an exact value.

c Compare the results of their calculations and explain what you notice.





11 A circle has a circumference of 52 cm.



- a Calculate the exact radius of the circle.
- b Hence, calculate the radius rounded to two decimal place.

12 Calculate the exact radius of the following circles:

- a A circle with circumference of 12π cm.
- b A circle with circumference of 12 cm.
- c A circle with circumference of 32π cm
- d A circle with circumference of 26 cm

13 Calculate the radius of the following circles, correct to two decimal places:

- a A circle with circumference of 14 cm.
- b A circle with circumference of 18 cm.
- c A circle with circumference of 14.9 cm.
- d A circle with circumference of 98 cm.

14 A circle has a circumference of 36 cm.

- a Calculate the exact diameter of the circle.
- b Hence, calculate the diameter rounded to two decimal place.

15 Calculate the exact diameter of the following circles:

- a A circle with circumference of 18π cm.
- b A circle with circumference 18 cm.
- c A circle with circumference of 17π cm.
- d A circle with circumference of 14 cm.

16 Calculate the diameter of the following circles, correct to two decimal places:

- a A circle with circumference of 37 cm.
- b A circle with circumference of 44 cm.
- c A circle with circumference of 70 cm.
- d A circle with circumference of 24.5 cm.

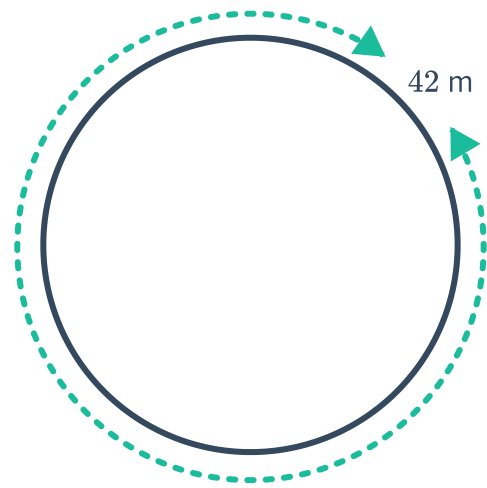
17 Complete the following table that lists the exact radii, diameters, and circumferences of various circles:

	Radius (cm)	Diameter (cm)	Circumference (cm)
Circle 1	2	4	
Circle 2	8		
Circle 3		82	
Circle 4			340

18 Consider the given circle:

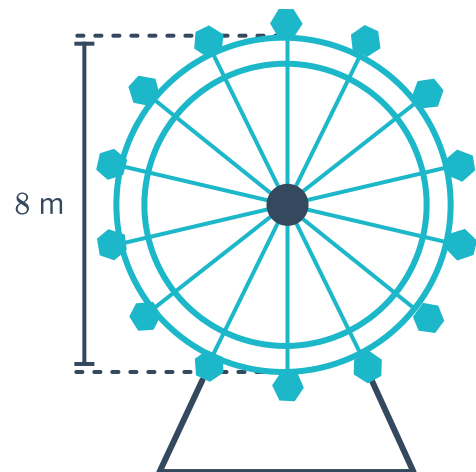


- a Find the diameter D , correct to two decimal places.
- b Find the radius r , correct to two decimal places.

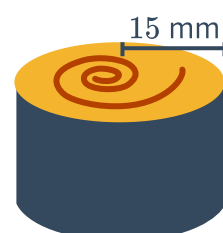


Applications

- 19 The bottom of a flower pot has a radius of 16 cm. Find the circumference of the bottom of the flower pot, correct to one decimal place.
- 20 A scooter tyre has a diameter of 34 cm. Find the circumference of the tyre, correct to one decimal place.
- 21 Find the circumference of the Ferris wheel, correct to one decimal place.



- 22 Find the length of the strip of seaweed around the outside of the sushi shown. Round your answer correct to one decimal place.



- 23** A coin has a diameter of 2.2 cm. If the coin is rolled through 45 complete revolutions, find the distance it will travel. Round your answer to one decimal place.
- 24** A circular running track has a diameter of 23 m. How many laps must be completed to run 1600 m? Round your answer to one decimal place.
- 25** The diameter of the front wheel of a car is 0.4 m. How many complete revolutions will the wheel make if the car travels a distance of 50 km?
- 26** The wheel of Roxanne's bicycle has a radius of 22 cm. How far would she cycle if the wheels made 250 complete revolutions? Give your answer in metres, correct to one decimal place.
- 27** Carl is performing an experiment by spinning a metal weight around on the end of a nylon thread. How far does the metal weight travel if it completes 40 revolutions on the end of a 0.65 m thread? Round your answer to one decimal place.
- 28** A shallow circular wading pool has a diameter of 4 m. A deeper circular swimming pool has a radius of 16 m. How many times greater is the circumference of the swimming pool than the wading pool?
- 29** A record has a diameter of 30 cm and is played at a speed of $33\frac{1}{3}$ revolutions per minute. Through what total distance will a point on the rim of the record travel if the record takes 28 minutes to play? Give your answer in metres, correct to one decimal place.
- 30** A satellite is orbiting the Earth at a height of h km above the Earth's surface. In one complete orbit, the satellite travels a distance of 41 664 km. If the radius of the Earth is 6400 km, find the height of the satellite above the Earth. Round your answer to one decimal place.

